

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q18-1046

Commonwealth Chesapeake 345 kV
September 29, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1046. The Interconnection Customer proposes a 287.1 MW natural gas combined cycle facility interconnecting at the Commonwealth Chesapeake 345 kV substation in the Virginia area. The request was received on August 26, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 412 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$27,199,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Ninemile Pt / Donner 138 kV	Cady / Dodd Road 345 kV	178.2	233	7.4
North Lebanon / Ninemile Pt 138 kV Ckt 1	Cady Substation / Dodd Rd 345 kV	163.2	365	14.9
Ninemile Pt / N Lebanon 138 kV L	Cady / Dodd Road 345 kV L	163.2	365	14.9
Seneca / Banta Carbona 138 kV	Cady / DoddRd 345 kV L	155.5	349	14.9
Quince Orchard / Colcheter 138 kV	Southtown / Commonwealth Chesapeake 345 kV	154.7	556	28.9
Persimmon Gap / Memphis Junction 138 kV	Vienna Operations / Easton 2 345 kV Line	154.2	409	20.2
Cady / Easton 2 345 kV	Easton 2 / Navy Mole 138 kV	154.1	1,716	69.5
Churchs Ferry / Willow Pass 138 kV	Cady / Dodd Rd 345 kV	153.8	196	8.8
Southtown / Commonwealth Chesapeake 345 kV	Huntington Falls / Otay Lake 138 kV	153.7	1,871	91.6
Mckee Run / Seneca 138 kV	Cady / Dodd Road Substation 345 kV	153.5	398	14.9
Saddlebrooke Ranch / Sunset 138 kV Line	Vienna Operations / Easton 2 Substation 345 kV	153.2	925	41.8
Tasey 345 kV/138 kV Autotransformer	Southtown / Commonwealth Chesapeake 345 kV L	153.1	608	24.0
Valleview / Colchester 138 kV	Southtown / Commonwealth Chesapeake 345 kV	151.1	587	28.9
Tulucay / Michoud Sta 138 kV	Commonwealth Chesapeake / Southtown 345 kV	149.7	167	8.4

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Prunedale / Dane 138 kV	Southtown / Commonwealth Chesapeake 345 kV	149.4	303	14.3
Huntington Falls / Commonwealth Chesapeake 138 kV	Commonwealth Chesapeake / Southtown 345 kV	149.1	748	34.6
Las Lomas / Newmark 138 kV	Vienna Operations / Easton 2 345 kV Line	149.0	260	10.2
Laglois Substation / Southtown 138 kV	Southtown / Commonwealth Chesapeake 345 kV Line	148.0	520	27.3
Otay Lake / Vedder 138 kV	Southtown / Commonwealth Chesapeake 345 kV	147.7	1,040	46.9
Quinn / Bulverde 138 kV Line	Vienna Operations Substation / Easton 2 345 kV	147.2	406	16.8
Huntington Falls / Otay Lake 138 kV	Commonwealth Chesapeake / Southtown 345 kV	147.2	1,048	46.9
Ninemile Pt / Longmire 138 kV L	Dodd Rd / Cady 345 kV	146.5	404	15.4
Ripple Substation / Mesquite 138 kV L	Southtown Substation / Vienna Operations Substation 345 kV	145.9	1,071	45.2
Longmire / Lost Hills 138 kV L	Cady Substation / Dodd Rd Substation 345 kV	145.4	440	16.1
Warren F Sam Beasley Generation Sta / Cady 138 kV	Cady / Dodd Road 345 kV	145.2	265	9.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Ninemile Pt / Donner 138 kV	Rebuild line with higher-capacity conductor	\$2,533,000	7% (\$168,000)
Ninemile Pt / N Lebanon 138 kV Cir. 1	Rebuild line with higher-capacity conductor	\$1,302,000	6% (\$80,000)
Ninemile Pt / N Lebanon 138 kV L	Rebuild line with higher-capacity conductor	\$1,749,000	6% (\$108,000)
Banta Carbona / Seneca 138 kV	Rebuild line with higher-capacity conductor	\$5,741,000	6% (\$348,000)
Quince Orchard / Colcheter 138 kV	Rebuild line with higher-capacity conductor	\$2,042,000	9% (\$190,000)
Persimmon Gap / Memphis Junction 138 kV	Rebuild line with higher-capacity conductor	\$2,733,000	7% (\$185,000)
Cady / Easton 2 345 kV	Rebuild line with higher-capacity conductor	\$114,438,000	6% (\$7,372,000)
Willow Pass / Churchs Ferry 138 kV	Rebuild line with higher-capacity conductor	\$2,755,000	6% (\$163,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Commonwealth Chesapeake / Southtown 345 kV	Rebuild line with higher-capacity conductor	\$47,957,000	9% (\$4,514,000)
Seneca / Mckee Run 138 kV	Rebuild line with higher-capacity conductor	\$3,960,000	6% (\$240,000)
Saddlebrooke Ranch / Sunset 138 kV L	Rebuild line with higher-capacity conductor	\$9,627,000	7% (\$663,000)
Tasey 345 kV/138 kV Autotransformer	Replace transformer with higher-rated unit	\$17,935,000	7% (\$1,281,000)
Valleview / Colchester 138 kV	Rebuild line with higher-capacity conductor	\$13,409,000	9% (\$1,251,000)
Tulucay / Michoud Sta 138 kV	Rebuild line with higher-capacity conductor	\$37,356,000	10% (\$3,680,000)
Prunedale / Dane Substation 138 kV	Rebuild line with higher-capacity conductor	\$3,100,000	9% (\$289,000)
Huntington Falls / Commonwealth Chesapeake 138 kV L	Rebuild line with higher-capacity conductor	\$3,483,000	9% (\$328,000)
Las Lomas / Newmark 138 kV L	Rebuild line with higher-capacity conductor	\$7,160,000	6% (\$457,000)
Langlois / Southtown 138 kV	Rebuild line with higher-capacity conductor	\$4,232,000	9% (\$368,000)
Otay Lake / Vedder 138 kV	Rebuild line with higher-capacity conductor	\$5,118,000	9% (\$473,000)
Quinn / Bulverde Substation 138 kV line	Rebuild line with higher-capacity conductor	\$830,000	7% (\$60,000)
Huntington Falls / Otay Lake 138 kV Line	Rebuild line with higher-capacity conductor	\$7,060,000	9% (\$653,000)
Ninemile Pt Substation / Longmire 138 kV	Rebuild line with higher-capacity conductor	\$10,152,000	6% (\$647,000)
Mesquite / Rippe 138 kV	Rebuild line with higher-capacity conductor	\$4,940,000	8% (\$383,000)
Longmire / Lost Hills 138 kV L	Rebuild line with higher-capacity conductor	\$15,401,000	6% (\$993,000)
Cady / Warren F Sam Beasley Generation Sta 138 kV	Rebuild line with higher-capacity conductor	\$36,293,000	6% (\$2,305,000)
	Total allocated to project		\$27,199,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.